以下是重新生成的 20 道题及选项：

1. 单选题（1 分）
The difference between ribose and deoxyribose is ().
A. The presence of an additional nitrogenous base in ribose
B. The presence of a hydroxyl group on the 3' carbon in deoxyribose
C. The absence of a nitrogenous base in deoxyribose
D. The absence of an oxygen atom on the 2' carbon in deoxyribose
E. The presence of an additional phosphate group in ribose

2. 单选题（1 分）
Which type of RNA carries amino acids to the ribosome ()?
A. mRNA
B. miRNA
C. tRNA
D. snRNA
E. rRNA

3. 单选题（1 分）
Which form of RNA carries the genetic code from DNA to the ribosome ()?
A. siRNA
B. tRNA
C. miRNA
D. rRNA
E. mRNA

4. 单选题（1 分）
The melting temperature () of DNA increases with ().
A. Higher thymine content
B. Higher guanine-cytosine content
C. Higher uracil content
D. Higher adenine content
E. Lower cytosine content

5. 单选题（1 分）
In which part of the cell does DNA replication occur ()?
A. Nucleus
B. Ribosome
C. Cytoplasm
D. Endoplasmic reticulum
E. Golgi apparatus

6. 单选题（1 分）
The RNA sequence complementary to the DNA sequence 5'-ATGCGT-3' is: ()

A. 5'-ACGCAU-3'
B. 5'-UAGCGU-3'
C. 5'-UGCGCA-3'
D. 5'-TACGCA-3'
E. 5'-AUCGCG-3'

7. 单选题（1 分）
Which RNA type forms the core of ribosome's structure ()?
A. tRNA
B. mRNA
C. siRNA
D. rRNA
E. snRNA

8. 单选题（1 分）
Which of the following base pairings is correct in DNA ()?
A. Guanine-Cytosine
B. Adenine-Uracil
C. Thymine-Uracil
D. Cytosine-Uracil
E. Adenine-Guanine

9. 单选题（1 分）
Which of the following is true about DNA and RNA ()?
A. Both are used for protein synthesis
B. Both contain deoxyribose sugar
C. Both are double-stranded
D. Both contain thymine
E. Both contain adenine, guanine, and cytosine

10. 单选题（1 分）
The phosphodiester bonds in DNA are formed between ().
A. The 1' hydroxyl group of one pentose and the 4' phosphate group of another
B. The phosphate group and the nitrogenous base
C. The nitrogenous base and the pentose
D. The 2' hydroxyl group of one pentose and the 3' phosphate group of another
E. The 3' hydroxyl group of one pentose and the 5' phosphate group of another

11. 单选题（1 分）
Which type of bond stabilizes the base pairs in the DNA double helix ()?
A. Peptide bond
B. Hydrogen bond
C. Ionic bond
D. Phosphodiester bond

E. Covalent bond

12. 单选题（1 分）
What is the primary function of tRNA during protein synthesis ()?
A. Transfer amino acids to the ribosome
B. Catalyze peptide bond formation
C. Carry genetic information
D. Unwind the DNA double helix
E. Serve as the template for mRNA

13. 单选题（1 分）
Which of the following processes involves the removal of introns ()?
A. Translation
B. Protein folding
C. DNA replication
D. Transcription
E. RNA splicing

14. 单选题（1 分）
Which of the following pairs represents a complementary base pairing in DNA ()?
A. Cytosine-Thymine
B. Adenine-Guanine
C. Uracil-Guanine
D. Thymine-Guanine
E. Guanine-Cytosine

15. 单选题（1 分）
Which component is unique to RNA but not found in DNA ()?
A. Deoxyribose
B. Ribose
C. Guanine
D. Cytosine
E. Phosphate

16. 单选题（1 分）
A mutation in which region would most likely affect protein translation ()?
A. Coding region of mRNA
B. Telomere
C. Promoter
D. Exon-intron boundary
E. Origin of replication

17. 单选题（1 分）
What stabilizes the double helix structure of DNA ()?

A. Disulfide bonds
B. Peptide bonds
C. Glycosidic bonds between sugars and bases
D. Ionic bonds between phosphate groups
E. Hydrogen bonds between the base pairs

18. 单选题（1 分）
What is the function of the 5' cap added to mRNA during processing in eukaryotes ()?
A. To protect mRNA from degradation
B. To signal the termination of translation
C. To suppress ribosome binding
D. To aid in DNA repair
E. To inhibit mRNA splicing

19. 单选题（1 分）
During transcription, what is synthesized from a DNA template ()?
A. Lipids
B. DNA
C. Proteins
D. RNA
E. Amino acids

20. 单选题（1 分）
Which sequence correctly describes the flow of genetic information ()?
A. Protein → RNA → DNA
B. RNA → DNA → Protein
C. DNA → RNA → Protein
D. Protein → DNA → RNA
E. DNA → Protein → RNA